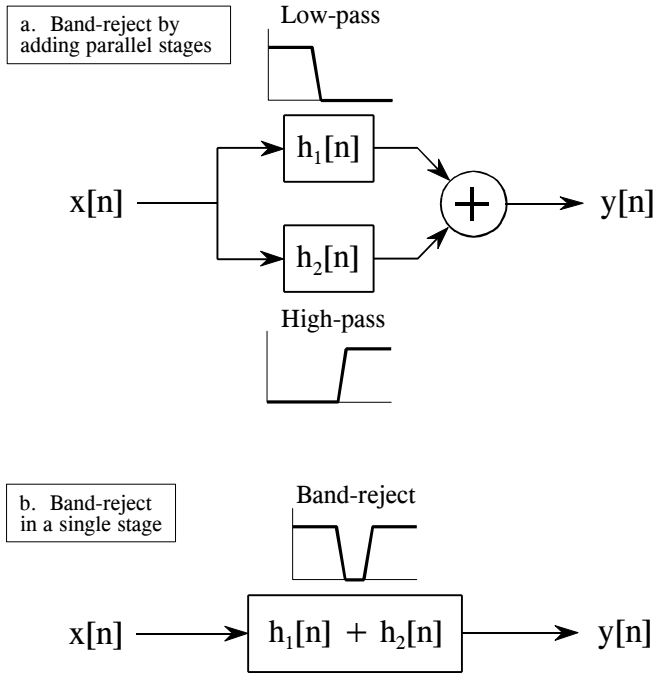

FIGURE 14-9
Designing a band-reject filter. As shown in (a), a band-reject filter is formed by the parallel combination of a low-pass filter and a high-pass filter with their outputs added. Figure (b) shows this reduced to a single stage, with the filter kernel found by *adding* the low-pass and high-pass filter kernels.

amplitude, frequency, and phase of the component sinusoids. The goal of these filters is to separate one band of frequencies from another. Custom filters are used when a special action is required by the filter, something more elaborate than the four basic responses (high-pass, low-pass, band-pass and band-reject). For instance, Chapter 17 describes how custom filters can be used for *deconvolution*, a way of counteracting an unwanted convolution.

		FILTER IMPLEMENTED BY:	
		Convolution *Finite Impulse Response (FIR)*	Recursion *Infinite Impulse Response (IIR)*
FILTER USED FOR:	Time Domain *(smoothing, DC removal)*	Moving average (Ch. 15)	Single pole (Ch. 19)
	Frequency Domain *(separating frequencies)*	Windowed-sinc (Ch. 16)	Chebyshev (Ch. 20)
	Custom *(Deconvolution)*	FIR custom (Ch. 17)	Iterative design (Ch. 26)

TABLE 14-1
Filter classification. Filters can be divided by their *use*, and how they are *implemented*.